

NARAYANA
IIT - JEE ACADEMY - INDIA

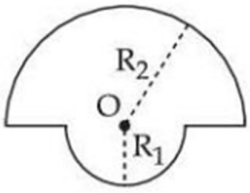
SR.IIT_*CO-SC(MODEL-A& APEX) **SPL GTM-06 (AN)** **Date: 19.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi m$ and $R_2 = 4\pi m$, carrying current $I = 4A$ as per figure given below is $\alpha \times 10^{-7} T$. The value of α is _____. (Centre O is common for all segments)



A) 1 B)2 C) 3 D) 5

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2. The threshold frequency of a metal with work function 6.63 eV is :

A) 1.6×10^{15} Hz B) 1.6×10^{12} Hz C) 16×10^{12} Hz D) 16×10^{15} Hz

3. The electric potential at the surface of an atomic nucleus $(z=50)$ of radius 9×10^{-13} cm is _____ $\times10^{6}$ V .

A) 2 B) 4 C) 6 D) 8

4. Primary side of a transformer is connected to 230V, 50Hz supply. Turns ratio of primary to secondary winding is 10 : 1. Load resistance connected to secondary side is 46Ω. The power consumed in it is:

A) 10.0W B) 11.5W C) 12.0 W D) 12.5 W

5. Given below are two statements :

Statement (I) : The limiting force of static friction depends on the area of contact and independent of materials.

Statement (II) : The force of kinetic friction is independent of the area of contact and depends on materials.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Both Statement I and Statement II are incorrect

B) Both Statement I and Statement II are correct

C) Statement I is incorrect but Statement II is correct

D) Statement I is correct but Statement II is incorrect

6. When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of :

A) 45° B) 30° C) 90° D) 60°

7. A current of 200μA deflects the coil of a moving coil galvanometer through 60°. The current to cause deflection through $\frac{\pi}{10}$ radian is

A) 180μA B) 120μA C) 60μA D) 30μA

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8. The total kinetic energy of 1 mole of oxygen at 27°C is :
[Use universal gas constant (R) = 8.31 J/mole K]
A) 6845.5J B) 5942.0J C) 5670.5J D) 6232.5J

9. A ball suspended by a thread swings in a vertical plane so that its magnitude of acceleration in the extreme position and lowest position are equal. The angle (θ) of thread deflection in the extreme position will be :
A) $2\tan^{-1}\left(\frac{1}{2}\right)$ B) $2\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$ C) $\tan^{-1}\left(\frac{1}{2}\right)$ D) $\tan^{-1}(\sqrt{2})$

10. The atomic mass of ${}_6C^{12}$ is 12.000000 u and that of ${}_6C^{13}$ is 13.003354 u. The required energy to remove a neutron from ${}_6C^{13}$, if mass of neutron is 1.008665 u, will be :
A) 49.5 MeV B) 4.95 MeV C) 62.5 MeV D) 6.25 MeV

11. Two charges of $-4\mu C$ and $+4\mu C$ are placed at the points $A(1,0,4)m$ and $B(2,-1,5)m$ located in an electric field $\vec{E}=0.20\hat{i}V/cm$. The magnitude of the torque acting on the dipole is
A) $8\times10^{-5}Nm$ B) $8\sqrt{5}\times10^{-5}Nm$ C) $8\sqrt{3}\times10^{-5}Nm$ D) $8\sqrt{2}\times10^{-5}Nm$

12. A closed organ pipe 150 cm long gives 7 beats per second with an open organ pipe of length 350 cm, both vibrating in fundamental mode. The velocity of sound is _____ m/s.
A) 300 B) 280 C) 294 D) 314

13. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio of $\frac{C_P}{C_V}$ for the gas is
A) $\frac{9}{7}$ B) $\frac{7}{5}$ C) $\frac{5}{3}$ D) $\frac{3}{2}$

14. A parallel beam of monochromatic light of wavelength 5000Å is incident normally on a single narrow slit of width 0.001mm. The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction

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given by

A) $\frac{\pi}{3}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{5}$ D) $\frac{\pi}{6}$

15. A heavy iron bar of weight 12 kg is having its one end on the ground and the other on the shoulder of a man. The rod makes an angle 60° with the horizontal, the weight experienced by the man is

A) 6 kg B) 3 kg C) $6\sqrt{3}$ kg D) 12 kg

16. The equation of state of a real gas is given by $\left(P+\frac{a}{V^2}\right)(V-b)=RT$, where P, V and T are pressure, volume and temperature respectively and R is the universal gas constant. The dimensions of $\frac{a}{b^2}$ is similar to that of :

A) PV B) RT C) R D) P

17. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The angular speed of the moon in its orbit about the earth is more than the angular speed of the earth in its orbit about the sun.

Reason (R) : The moon takes less time to move around the earth than the time taken by the earth to move around the sun.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

B) (A) is not correct but (R) is correct

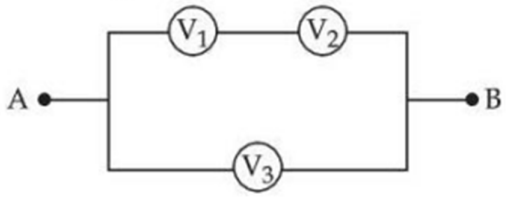
C) (A) is correct but (R) is not correct

D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

18. Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.

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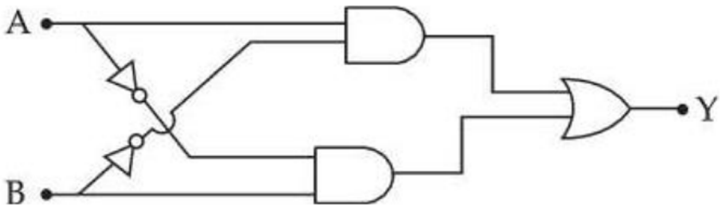


A) $V_1 \neq V_3 - V_2$ B) $V_1 = V_2$ C) $V_1 + V_2 = V_3$ D) $V_1 + V_2 > V_3$

19. Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L, radius r. If X is the resistance of wire, then specific resistance is; $S_1 = X\left(\frac{\pi r^2}{L}\right)$. If the length of the wire gets double then the value of specific resistance will be :

A) $\frac{S_1}{4}$ B) $2S_1$ C) $\frac{S_1}{2}$ D) S_1

20. The truth table of the given circuit diagram is :



A)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

B)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

C)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A body falling under gravity covers two points A and B separated by 80 m in 2 s. The distance of upper point A from the starting point is _____ m (use g = 10 ms^{-2}).

22. A series LCR circuit with $L=\frac{100}{\pi}mH$, $C=\frac{10^{-3}}{\pi}F$ and $R=10\Omega$, is connected across an ac source of 220 V, 50 Hz supply. The power factor of the circuit would be _____.

23. The reading of pressure metre attached with a closed pipe is $4.5\times10^4 N/m^2$. On opening the valve, water starts flowing and the reading of pressure metre falls to $2.0\times10^4 N/m^2$. The velocity of water is found to be $\sqrt{V}m/s$. The value of V is _____.

24. If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where $\alpha=$ _____.

25. A ring and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of both bodies are identical and the ratio of the their kinetic energies is $\frac{7}{x}$, where x is _____.

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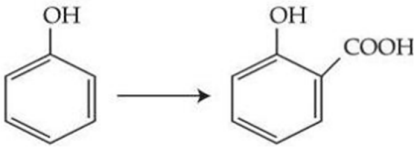
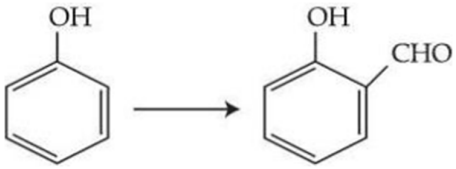
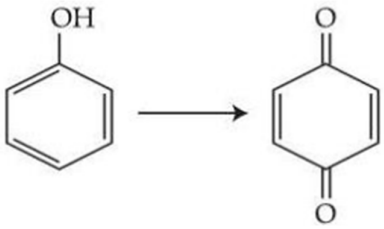
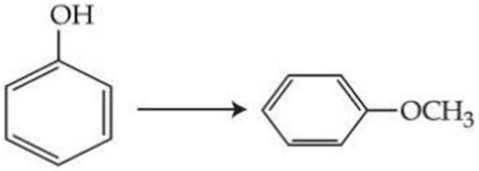
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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Match List – I with List – II.

List – I (Reaction)	**List – II (Reagent(s))**
(a) 	(I) $Na_2Cr_2O_7, H_2SO_4$
(b) 	(II) (i) NaOH (ii) CH_3Cl
(c) 	(III) (i) $NaOH, CHCl_3$ (ii)NaOH (iii) HCl
(d) 	(IV) (i) NaOH (ii) CO_2 (iii) HCl

Choose the correct answer from option below.

A) (a)-(IV), (b)-(I), (c)-(III), (d)-(II)
B) (a)-(IV), (b)-(III), (c)-(I), (d)-(II)
C) (a)-(II), (b)-(I), (c)-(III), (d)-(IV)
D) (a)-(II), (b)-(III), (c)-(I), (d)-(IV)

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27. Which structure of protein remains intact after coagulation of egg white on boiling?

A) Tertiary B) Secondary C) Primary D) Quaternary

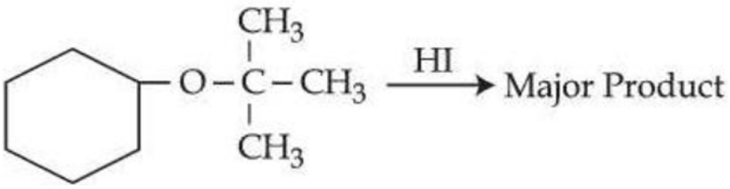
28. Which of the following cannot function as an oxidizing agent?

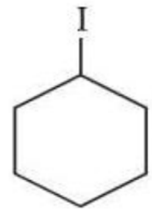
A) MnO_4^- B) SO_4^{2-} C) BrO_3^- D) N^{3-}

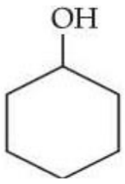
29. The molecular formula of second homologue in the homologous series of mono carboxylic acid is ___.

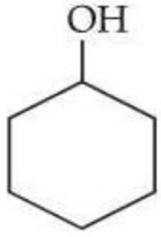
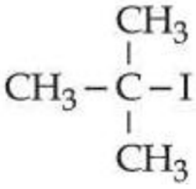
A) $C_2H_2O_2$ B) $C_3H_6O_2$ C) $C_2H_4O_2$ D) CH_2O

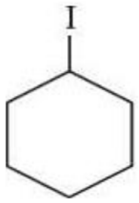
30. Major product formed in the following reaction is a mixture of :



A)  and $(CH_3)_3CI$

B)  and $(CH_3)_3COH$

C)  and 

D)  and $(CH_3)_3COH$

31. Identify B formed in the reaction.

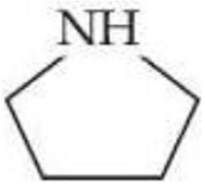
$$Cl-(CH_2)_4-Cl \xrightarrow{\text{excess } NH_3} A \xrightarrow{NaOH} B + H_2O + NaCl$$

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A) $H_2N-(CH_2)_4-NH_2$

B) 

C) 

D) $\overset{+}{Cl}NH_3-(CH_2)_4-\overset{+}{N}H_3Cl^-$

32. Phenolic group can be identified by a positive :

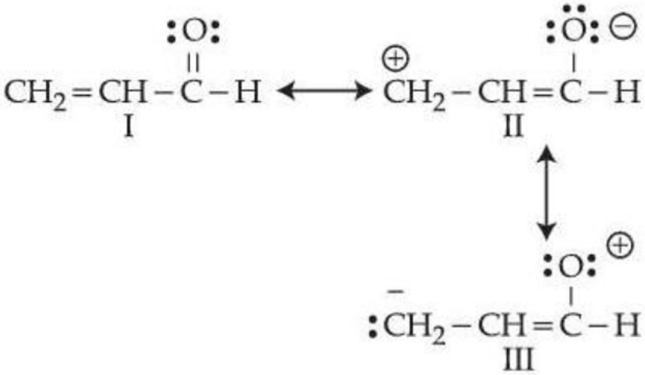
A) Lucas test B) Tollen's test

C) Phthalein dye test D) Carbylamine test

33. The quantity which changes with temperature is :

A) Mole fraction B) Molality C) Molarity D) Mass percentage

34. The order of relative stability of the contributing structure is :



Choose the correct answer from the options given below :

A) III > II > I B) I > II > III C) II > I > III D) I = II = III

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35. The technique used for purification of steam volatile water immiscible substance is :

A) Fractional distillation under reduced pressure

B) Steam distillation

C) Distillation

D) Fractional distillation

36. Identify from the following species in which d^2sp^3 hybridization is shown by central atom

A) SF_6 B) $[Pt(Cl_4)]^{2-}$ C) BrF_5 D) $[Co(NH_3)_6]^{3+}$

37. Which of the following statements is not correct about rusting of iron?

A) Coating of iron surface by tin prevents rusting, even if the tin coating is peeling off.

B) When pH lies above 9 or 10, rusting of iron does not take place

C) Dissolved acidic oxides SO_2, NO_2 in water act as catalyst in the process of rusting

D) Rusting of iron is envisaged as setting up of electrochemical cell on the surface of iron object

38. Given below are two statements :

Statement (I) : In the Lanthanoids, the formation Ce^{+4} is favoured by its noble gas configuration.

Statement (II) : Ce^{+4} is a strong oxidant reverting to the common + 3 state.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

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39. Given below are two statements :

Statement (I) : Oxygen being the first member of group 16, exhibits only –2 oxidation state.

Statement (II) : Down the group 16, stability of + 4 oxidation state decreases and + 6 oxidation state increases.

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Both Statement I and Statement II are correct

B) Statement I is correct but Statement II is incorrect

C) Statement I is incorrect but Statement II is correct

D) Both Statement I and Statement II are incorrect

40. The final product A, formed in the following reaction sequence is :

$$Ph-CH=CH_2 \xrightarrow[(iii)\ HBr]{\begin{array}{l}(i)\ BH_3\\(ii)\ H_2O_2, \ominus OH\end{array}} A \xrightarrow{(iv)\ Mg,\ ether,\ then\ HCHO/H_3O^+}$$

A) $Ph-\underset{\displaystyle CH_2OH}{\underset{|}{CH}}-CH_3$

B) $Ph-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

C) $Ph-CH_2-CH_2-CH_2-OH$

D) $Ph-CH_2-CH_2-CH_3$

41. The incorrect statement regarding conformations of ethane is

A) Eclipsed conformation is the most stable conformation

B) Ethane has infinite number of conformations

C) The conformations of ethane are inter-convertible to one-another

D) The dihedral angle in staggered conformation is 60°

42. Identify the incorrect pair from the following :

A) Haber process – Iron

B) Polythene preparation - $TiCl_4, Al(CH_3)_3$

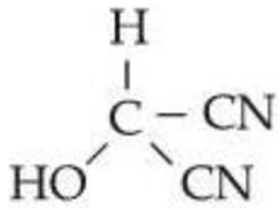
C) Wacker process - $PtCl_2$

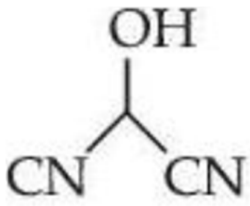
D) Photography - $AgBr$

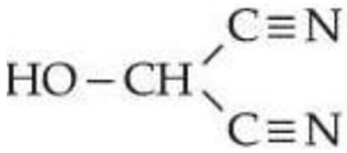
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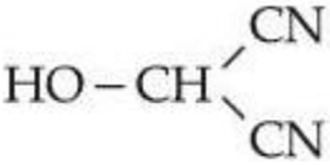
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43. Bond line formula of $HOCH(CN)_2$ is :

A) 

B) 

C) 

D) 

44. Choose the correct option having all the elements with d^{10} electronic configuration from the following:

A) $_{29}Cu, _{30}Zn, _{48}Cd, _{47}Ag$

B) $_{28}Ni, _{24}Cr, _{26}Fe, _{29}Cu$

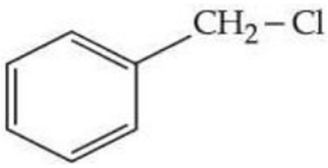
C) $_{46}Pd, _{28}Ni, _{26}Fe, _{24}Cr$

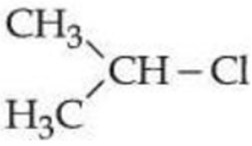
D) $_{27}Co, _{28}Ni, _{26}Fe, _{24}Cr$

45. Which among the following halide/s will not show S_N1 reaction :

(a) $H_2C=CH-CH_2Cl$

(b) $CH_3-CH=CH-Cl$

(c) 

(d) 

Choose the most appropriate answer from the options given below :

A) (a), (b) and (d) only

B) (a) and (b) only

C) (b) only

D) (b) and (c) only

APEX JEE-MAIN 2025

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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46. The number of non-polar molecules from the following is _____.

$HF, H_2O, SO_2, H_2, CO_2, CH_4, NH_3, HCl, CHCl_3, BF_3$

47. 9.3 g of aniline is subjected to reaction with excess of acetic anhydride to prepare acetanilide. The mass of acetanilide produced if the reaction is 100% completed is _____ $\times 10^{-1}$ g.

(Given molar mass in g mol^{-1} $N:14, O:16, C:12, H:1$).

48. For a certain thermochemical reaction $M \rightarrow N$ at $T = 400K$, $\Delta H^{\ominus} = 77.2$ kJ mol^{-1}, $\Delta S = 122$ JK^{-1}, log equilibrium constant (log K) is $-$ _____ $\times 10^{-1}$.

49. Time required for completion of 99.9% of a First order reaction is _____ times of half life $(t_{1/2})$ of the reaction.

50. The hydrogen electrode is dipped in a solution of $pH = 3$ at $25°C$. The potential of the electrode will be $-$_____ $\times 10^{-2}$V. $\left(\frac{2.303RT}{F} = 0.059V\right)$

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MATHS **MAX.MARKS: 100**

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in N$, then $\sum_{k=1}^{n} \frac{k}{2^k}$ is equal to

A) $\frac{1}{2^{13}}(2^{14}-15)$ B) $1-\frac{15}{2^{13}}$ C) $\frac{1}{2^{15}}(2^{14}-14)$ D) $\frac{1}{2^{14}}(2^{15}-15)$

52. The values of α, for which $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$, lie in the interval

A) (0, 3) B) $\left(-\frac{3}{2}, \frac{3}{2}\right)$ C) $(-2,1)$ D) (–3, 0)

53. For $0 < a < 1$, the value of the integral $\int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}$ is

A) $\frac{\pi^2}{\pi - a^2}$ B) $\frac{\pi}{1-a^2}$ C) $\frac{\pi}{1+a^2}$ D) $\frac{\pi^2}{\pi + a^2}$

54. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$ respectively. Let ℓ_1, ℓ_2 and ℓ_3 be the lengths of perpendiculars drawn from the ortho centre of the triangle on the sides AB, BC and CA respectively, then $\ell_1^2 + \ell_2^2 + \ell_3^2$ equals :

A) $\frac{1}{5}$ B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) $\frac{1}{3}$

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55. The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let l denotes the length of the angle bisector AD of $\angle BAC$ where D is on the line segment BC, then $2l^2$ equals :

A) 50 B) 42 C) 45 D) 49

56. Let R be the interior region between the lines $3x-y+1=0$ and $x+2y-5=0$ containing the origin. The set of all values of a, for which the points $(a^2, a+1)$ lie in R, is

A) $(-3,-1)\cup\left(-\frac{1}{3},1\right)$ B) $(-3,0)\cup\left(\frac{1}{3},1\right)$

C) $(-3,-1)\cup\left(\frac{1}{3},1\right)$ D) $(-3,0)\cup\left(\frac{2}{3},1\right)$

57. Consider only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$ is

A) 2 B) 1 C) 0 D) More than 2

58. Let the image of the point $(1,0,7)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α,β,γ). Then which one of the following points lies on the line passing through (α,β,γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis?

A) $(3,4,3-2\sqrt{2})$ B) $(3,-4,3+2\sqrt{2})$ C) $(1,-2,1+\sqrt{2})$ D) $(1,2,1-\sqrt{2})$

59. The 20th term from the end of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots, -129\frac{1}{4}$ is :

A) – 115 C) – 118 C) – 110 D) – 100

60. Let $g(x)=3f\left(\frac{x}{3}\right)+f(3-x)$ and $f''(x)>0$ for all $x\in(0,3)$. If g is decreasing in $(0,\alpha)$ and increasing in $(\alpha,3)$, then 8α is

A) 20 B) 24 C) 0 D) 18

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61. If $\lim\limits_{x\to 0}\dfrac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x}=\dfrac{1}{3}$, then $2\alpha-\beta$ is equal to

A) 5 B) 7 C) 1 D) 2

62. If $y=y(x)$ is the solution curve of the differential equation $(x^2-4)dy-(y^2-3y)dx=0$, $x>2$, $y(4)=\dfrac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals

A) $\dfrac{3}{1-2\sqrt{2}}$ B) $\dfrac{3}{1-(8)^{1/4}}$ C) $\dfrac{3}{1+2\sqrt{2}}$ D) $\dfrac{3}{1+(8)^{1/4}}$

63. If α,β are the roots of the equation, $x^2-x-1=0$ and $S_n=2023\alpha^n+2024\beta^n$, then

A) $2S_{12}=S_{11}+S_{10}$ B) $S_{11}=S_{10}+S_{12}$ C) $S_{12}=S_{11}+S_{10}$ D) $2S_{11}=S_{12}+S_{10}$

64. Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point P(m, n) from the point $Q(-2,-3)$ is :

A) 4 B) 6 C) 8 D) 10

65. Consider the function $f:(0,2)\to R$ defined by $f(x)=\dfrac{x}{2}+\dfrac{2}{x}$ and the function g(x) defined by

$$g(x)=\begin{cases}\min\{f(t)\}, & 0<t\le x \text{ and } 0<x\le 1\\ \dfrac{3}{2}+x, & 1<x<2\end{cases}$$, Then

A) g is neither continuous nor differentiable at x = 1

B) g is continuous and differentiable for all $x\in(0,2)$

C) g is continuous but not differentiable at x = 1

D) g is not continuous for all $x\in(0,2)$

66. Let $f:R-\left\{\dfrac{-1}{2}\right\}\to R$ and $g:R-\left\{\dfrac{-5}{2}\right\}\to R$ be defined as $f(x)=\dfrac{2x+3}{2x+1}$ and $g(x)=\dfrac{|x|+1}{2x+5}$. Then, the domain of the function fog is :

A) $R-\left\{-\dfrac{5}{2}\right\}$ B) $R-\left\{-\dfrac{5}{2},-\dfrac{7}{4}\right\}$ C) $R-\left\{-\dfrac{7}{4}\right\}$ D) R

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67. Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$. Then :

A) $\alpha \notin N$ and $\beta \notin N$ B) $\alpha \in N$ and $\beta \in N$ C) $\alpha \notin N$ and $\beta \in N$ D) $\alpha \in N$ and $\beta \notin N$

68. The integral $\int \frac{(x^8 - x^2)dx}{(x^{12} + 3x^6 + 1)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ is equal to

A) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right) + C$ B) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^3 + C$

C) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/2} + C$ D) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/3} + C$

69. Let e_1 be the eccentricity of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and e_2 be the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$, which passes through the foci of the hyperbola. If $e_1 e_2 = 1$, then the length of the chord of the ellipse parallel to the x-axis and passing through (0, 2) is

A) $4\sqrt{5}$ B) $\frac{8\sqrt{5}}{3}$ C) $\frac{10\sqrt{5}}{3}$ D) $3\sqrt{5}$

70. An urn contains 6 white and 9 black balls. Two successive draws of 4 balls are made without replacement. The probability, that the first draw gives all white balls and the second draw gives all black balls, is

A) $\frac{5}{715}$ B) $\frac{5}{256}$ C) $\frac{3}{256}$ D) $\frac{3}{715}$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$, where g is a continuous odd function. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx=\left(\frac{\pi}{\alpha}\right)^2-\alpha$, then α is equal to _____.

72. Consider a circle $(x-\alpha)^2+(y-\beta)^2=50$, where $\alpha,\beta>0$. If the circle touches the line $y+x=0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha+\beta)^2$ is equal to _____.

73. Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circle $|z-z_0|^2=4$ and $|z-z_0|^2=16$ respectively, where $z_0=1+i$. Then, the value of $100|\alpha|^2$ is _____.

74. Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A-xI|=0$ be –1 and 3, then the sum of the diagonal elements of the matrix A^2 is _____.

75. If the solution curve, of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point (2, 1) is $\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$, then $5\beta+\alpha$ is equal to

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